What a Repair Leaves Behind: Evaluating Meaning-Preservation Interventions for LLM Sentence Expansion in AAC with Benefit, Harm and Cost Together

Anonymous submission

Abstract

Augmentative and alternative communication (AAC) apps increasingly let a large language model expand a few selected words into a sentence spoken on the user's behalf. Proposed fixes for meaning errors are usually reported by their targeted gain alone. We evaluate three interventions with an offline design that reports benefit, harm and cost together and fixes a kill criterion for each in advance. The design attributes each error by construction to the input channel (cue never supplied vs. supplied and changed), so each intervention is scored only where it can act; it pairs every treated output with the stored untreated output for the same request; it resamples whole item families; and it keeps an explicit harm ledger. A runtime guard that checks supplied commitments and regenerates repaired 63/80 scored violations (78.8%) at 0.104 extra model calls per input, changed 2.3% of correct outputs, and turned none into a scored failure. It also left 17 violations in place and produced repairs that pass the check without being usable sentences. Filling missing readings into a three-suggestion set raised sets offering both polarity or both certainty readings from 15/80 to 78/80, at 2.05 calls per set; politeness coverage fell by 35.0 points and 32.2% of rewrites changed the sentence type. One user-selected cue raised critical-value match from 41.8% to 94.8% for one nominal selection. All three pass their kill criteria, and none comes without a measured cost. Labels are machine-extracted and human validation is pending; we claim no human effects. We close with a micro-randomized protocol that tests whether these offline ledgers predict causal, longitudinal effects for AAC users.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns want go party into a full sentence (Shen et al. 2022; Cai et al. 2024), and that sentence is then spoken as the user's own (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). The model chooses more than words. It also decides whether something happened or did not, who should act, and whether the speaker sounds certain or polite. When it chooses wrongly, the person says something they did not mean.

Several interventions could reduce this. A system can check its output against what the user supplied and try again (Madaan et al. 2023; Gou et al. 2024); it can offer suggestions that span the readings the keywords leave open; or the user can add a cue such as [NOT] (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025). Such interventions are usually reported by the gain on the endpoint they target. That is not enough for an AI intervention that speaks for a person. A repair can fix a dropped negation and also rewrite a sentence that was already right. A suggestion set can gain a missing reading by giving up one the user needed. Each fix costs model calls, latency or user effort. In AI-mediated communication, small shifts in suggested wording change what people write and how others see them (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Hohenstein et al. 2023; Zhang et al. 2026; Agarwal, Naaman, and Vashistha 2025).

We therefore evaluate interventions with a ledger instead of a score. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, built in concurrent work that specifies and tests meaning-preservation requirements for this task (Anonymous 2027). Here we ask a different question: *what does each intervention buy, what does it break, and what does it cost?*

Contributions. (1) An offline evaluation design for text-generation interventions, built from five parts: channel-controlled attribution, which decides which defect an intervention may be credited with; paired treated/untreated outputs; family-cluster bootstrap; an explicit harm ledger; and a kill criterion fixed before the test run (§5, Table 1). (2) An evaluation of two system-side interventions and one user-side intervention (§6, Figs. 1–2). All three pass their kill criteria. Each also has a documented failure mode that a gain-only report would hide. (3) A protocol for the human study these offline results cannot replace: a micro-randomized trial with causal excursion estimands, co-adaptation and carry-over outcomes, and stopping rules that mirror the offline ledger (§7).

We ran no study with AAC users and claim no effect on communication, agency, effort or trust. Every result below is a change in stored text-system behavior on constructed items.

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2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how such systems help and hinder them (Valencia et al. 2023). Several add user-side control over intent: context and intent choices (Valencia et al. 2024; Mao et al. 2025), editable stance tags (Xiang et al. 2025), and retained function words such as "not" (Lacosse, Tano, and McNamee 2026). Others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025; Gorenshtein et al. 2026). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026). Our ledger implements the technical layer in a form that the functional layer (§7) can test.

Self-correction and suggestion sets. Generate–critique–revise loops are established (Madaan et al. 2023), including loops with external tool feedback (Gou et al. 2024). We do not claim the loop. Our guard's critic is a deterministic monitor for commitments the user *supplied*, and we evaluate the loop's collateral effects, not only its fixes. Writing with language models can narrow the range of what people write (Padmakumar and He 2024; Arnold, Chauncey, and Gajos 2020), which is why we score whole suggestion sets across all meaning dimensions.

Evaluating AI-mediated communication. Hancock et al. define AI-mediated communication and call for studying its effects on self-presentation and relationships (Hancock, Naaman, and Levy 2020). Opinionated co-writers shift users' views (Jakesch et al. 2023), AI replies change language and social perception (Hohenstein et al. 2023), suggested politeness strategies change both email wording and how recipients judge the writer (Zhang et al. 2026), and suggestions pull writing toward dominant styles (Agarwal, Naaman, and Vashistha 2025). These are human-outcome studies of exposure; ours is an offline, pre-deployment study of interventions meant to *prevent* unwanted shifts, and §7 specifies how its effects would be tested causally. On measurement, behavioral test suites (Ribeiro et al. 2020) and LLM judges (Zheng et al. 2023) are common, as are their biases (Panickssery, Bowman, and Feng 2024); prediction-powered inference corrects automated estimates with a small human sample (Angelopoulos et al. 2023).

3 Setting

Items and channels. We reuse the frozen held-out test split of the concurrent study (Anonymous 2027). It contains minimal-pair *families* on six dimensions: polarity, who acts (agent), certainty, speech act, politeness, and time or quantity. The two source variants of a family differ only on the family's critical dimension ("I don't like the new medication" / "I like the new medication") and collapse to the same *bare* keywords (like medication). Each variant is rendered through five input *channels*: *bare*; *noncrit*, which adds one non-critical word as a length control (like new medication); *chip1*, which adds the critical commitment chip ([NOT] like medication); *chipall*, with chips for all commitments; and *full*, the typed source sentence. The primary population has 58 families after a pre-declared quarantine of two refusal/acceptance families whose scoring rule has a known blind spot (Anonymous 2027). Families and probes are constructed; they are not messages written by AAC users.

Attribution decides what an intervention may claim. Because the channel fixes which commitments the model received, every miss is attributed by construction. *Loss*: the channel never carried the value (bare, noncrit). *Violation*: the channel carried it and the output changed it. *Assumption*: the channel was ambiguous and the output committed anyway (Anonymous 2027). This attribution is the first step of our evaluation design. A system-side checker can only enforce a commitment the user actually supplied, so the guard is credited with fixing Violations alone. When the keywords leave a dimension open, no checker can know the intended reading, so the remedies are to show the alternatives (I2) or to let the user supply the cue (chip). Pooling all misses would credit the guard for errors it cannot see, and would blame the model for information the interface never collected.

Systems and scoring. The systems under test are Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit GGUF via llama.cpp) and the API model gpt-6.1-terra. P0 asks for one sentence, decoded greedily; P2 asks for three suggestions that "cover the different things the user might mean"; S3 draws three samples at temperature 0.7. An LLM extractor (gpt-6.1-sol, two prompt variants, cross-family arbiter on disagreement) labels each output *blind* to source, channel, model and arm. Deterministic code then computes verdicts from those labels. The extractor never sees which arm produced a text, so treated and untreated outputs are scored by the same blind procedure. The oracle's own validation (identity and mutation tests) is reported in the concurrent study (Anonymous 2027); human validation is pending (§8).

4 Interventions

I1: commitment guard (system side, runtime). After the first-pass P0 output, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the commitments supplied in the visible input. For chips these are the chips themselves; for full input they are the monitor's own labels of the typed source. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated up to twice. The guard returns the first unflagged attempt, else the attempt with fewest flags, with ties going to the original. The monitor never sees the evaluation extractor's labels. *G1s* is a zero-call comparator that only deletes an unsupplied "please".

I2: ambiguity-complete suggestion set (system side, bare input). Starting from the P2 set, I2 keeps the top suggestion. For polarity and then certainty, if the monitor finds no slot carrying the alternative reading, I2 rewrites the third

(then second) slot's own sentence into that reading with one LLM call ("make it UNCERTAIN: a statement that is not sure (use maybe, might or I think)"). The rewrite is verified by the monitor, with one feedback retry. A variant, I2t, rewrites the top suggestion instead; I2 was chosen over I2t on a 12-family development pilot before the test run, and both are reported.

Chip (user side). The user adds one commitment chip, such as [NOT], [MAYBE], [?] or [PLAIN], to the keywords. We evaluate it as an intervention on the same footing as the system-side ones: compared with a length-matched non-critical word, with a nominal cost of one selection.

5 Evaluation Design

Table 1 summarizes the design in the form the workshop asks for: intervention, comparison, population, outcomes and time horizon.

Paired, within-request comparisons. For the two system-side interventions, the untreated outcome is the *stored* first-pass output for the same request. The treated outcome is what the intervention returns from that output. Both potential outcomes are therefore observed for every unit, and the contrast is free of between-request variation. The pairing matters because greedy decoding is not reproducible even within one serving setup: issuing the same greedy request twice reproduced the same text in only 76.4–90.8% of local and 44.4–71.4% of API requests in follow-up runs of the same models (two greedy draws per request). A design that regenerated the control would mix intervention effect with this run-to-run churn. The chip comparison is paired across channels of the same target: chip1 against the length-matched noncrit word. Those are different requests, so for chips, run-to-run churn is part of the contrast.

Estimands and uncertainty. Each estimand is an average over the benchmark's targets or sets, weighted by row. Families are the resampling unit: every interval is a 95% percentile interval from 2,000 whole-family bootstrap resamples (seed 20261031), and paired arms, models, variants and channels stay together within a family. The four models are fixed, not sampled. Thousands of target rows are not thousands of independent tests: the guard's 63/80 repaired violations come from 26 families.

Harm ledger. For every intervention we report, beside its benefit: (i) *changed-correct*, the share of already-correct outputs whose text the intervention changed; (ii) *pass→fail*, the share that became scored failures; (iii) *residual*, violations or misses it left in place; (iv) *collateral coverage loss*, sets that offered both readings of a dimension the intervention did not target before it and do not after; and (v) *act drift*, rewrites that change the sentence type (statement, question, request). (i) is a harm even when (ii) is zero. An AAC user who typed "Would you mind repeating the last question?" gets a different sentence even if a coarse label still calls it polite. The ledger also records diagnostics a label cannot see: repairs that pass the scorer but are not sentences.

Cost. We count extra generator calls and wall-clock latency per input (I1) or per set (I2), and nominal selections for chips. Local latencies were measured with partial CPU offload, while another process held GPU memory, so they are upper bounds that are comparable only within this study.

Kill criteria. Each intervention had a pass/fail rule (Table 1) checked on a 12-family development pilot before the held-out run; a failing intervention would have been reported as killed, not tuned on test. They are coarse by design: they decide whether an intervention is worth a human study, not whether it is good.

6 Results

6.1 I1: the guard repairs most violations, and its harms are visible

On cue channels, strict pass (critical value kept, no supplied dimension contradicted, no politeness added) rose from 94.3% [91.9, 96.3] to 98.8% [97.9, 99.5] (Fig. 1, top). The guard repaired 63/80 scored violations, 78.8% [68.4, 88.9], far above its 30% kill line. The zero-call strip-only comparator repaired 42.5% [28.8, 54.4], all of them politeness additions. On ambiguous channels, where only the politeness rule applies, the guard removed all 86 unsupplied "please" additions. Including all channels, it repaired 89.8% of 166 violations.

Harm ledger. The guard changed the text of 49/2,154 already-correct outputs (2.3% [1.1, 3.6]), well inside its 10% kill line. Most of these changes fell on ambiguous channels: removing an unsupplied "please" from a bare output also edits that output for the polite variant, which it happened to match. On cue channels the guard changed 14/1,312. None became a scored failure (0/2,154). This does not mean they were harmless: the gallery's changed-correct row (Fig. 2) turns "Would you mind repeating the last question" into "Please repeat the last question", which is still polite by the label and is still not what was typed. 17 of 80 violations remained. In the gallery, the guard regenerates "Can you charge my wheelchair?" for [?] [YOU] charge wheelchair and returns the same request. Of the residual failures, 12/17 involve speech act, and 14/17 were still flagged by the monitor: they were detected but not repaired, not invisible. Some repairs pass the scorer without being sentences: 4/63 begin with the chip echo "Ask you", as in "Ask you to wear headphones." for [ASK] [YOU] [PLAIN] wear headphones. This count is a post hoc lexical diagnostic, not a fluency rate; no fluency or acceptability endpoint was measured. A blind human audit of 20 repaired and 20 unchanged outputs is planned before any usability claim.[PENDING: repair audit]

Cost and dependence. The guard flagged 157/1,856 inputs (8.5%) and added 0.104 generator calls per input (1.23 per flagged input) and 0.17 s mean latency per input (median 1.59 s per flagged input); the monitor check takes 2.3 ms. Monitor and evaluator are independent programs but measure the same constructs. On first-pass cue outputs the monitor's flag agreed with judged failure at $\kappa = 0.87$ (precision 0.80, recall 0.96). A guard "fix" is therefore partly a fix as

	I1 commitment guard	I2 ambiguity-complete set	Chip (user side)
Defect targeted	Violation of a supplied commitment	Assumption under ambiguous input	Loss of a cue at input
Comparison	Stored P0 first pass, same request; G1s strip-only (0 calls)	Stored P2 set, same request; S3 samples; I2t variant	Length-matched non-critical word (noncrit), same target
Unit, population	(model, family, variant, cue channel) target; 1,392 targets, 58 families	(model, family) bare-input set of 3; one set per family per model	(model, family, variant) target; 464 targets
Primary benefit	Share of scored violations repaired	Sets offering both readings, polarity and certainty families	Critical-value match, chip1 vs. noncrit
Harm ledger	Correct outputs changed; correct→failure; violations left; unusable repairs	Coverage lost on non-targeted dimensions; sentence-type (act) drift of rewrites	Matches lost when the chip is added; misses not repaired; contradictions
Cost	Generator calls, latency, monitor time	Generator calls, latency	Nominal selections (reading time not counted)
Kill criterion	Repair $\geq$30% of violations *and* change $\leq$10% of correct outputs	Gain $\geq$30 points over P2	Gain $\geq$10 points over noncrit *and* $\geq$80% of full-input match
Human outcome (planned)	Message sent as meant; corrections; time to send	Suggestion accepted; meaning endorsed	Chip use; message sent as meant
Time horizon (planned)	Per message; weeks (reliance, co-adaptation)	Per set; weeks (style drift)	Per message; weeks (habituation)

Table 1: Evaluation card. Rows above the rule "Kill criterion" are evaluated offline here; the last two rows are planned for the human study (§7) and nothing about them is measured. Kill criteria were fixed in the analysis code and the study plan and checked on development data before the held-out run; they were not externally timestamped.

seen by an extractor of the same kind, which is why the human audit above matters. Including the two quarantined families, the guard repaired 67/93 (72.0% [59.6, 84.2]); this sensitivity has a different population and denominator and is not pooled with the primary estimate. For explicitly typed full sentences, returning the input verbatim passes the same contract in 116/116 stored source identities at zero calls. This baseline needs no model, and it is the right default in a declared full-sentence mode.

6.2 I2: completing a set displaces other readings

Under bare input, the P2 set offered both polarity or both certainty readings in 15/80 sets (18.8% [10.0, 28.7]); I2 raised this to 78/80 (97.5% [93.8, 100.0]), a paired gain of 78.8 points [70.0, 87.5], far above its kill line. Sampling three times (S3) offered both readings in 1.2%. I2 gained coverage in 63/65 previously uncovered sets and lost none.

Harm ledger. The slots I2 rewrites are the slots that carried other readings. Politeness coverage fell from 26/40 to 12/40 sets, a loss of 35.0 points [15.0, 57.5], and speech-act coverage fell by 25.8 points [9.7, 40.6]. Across the non-targeted dimensions, I2 removed both-reading coverage from 25/62 sets that had it and added it to 2/85 that did not. Coverage over all six dimensions still rose by 17.6 points [4.0, 31.6], so the net is positive, but the dimensions that pay are different from the ones that gain. The gallery's loss row makes this concrete: for knock come, the polite request "Please knock before you come in." is rewritten into "Don't knock before you come in." Rewrites also change sentence type. Of 428 rewrites across 58 families, 138/428 (32.2% [28.2, 36.3]) change speech act; 129 of these end as statements. Certainty rewrites change act more often than polarity rewrites (96/220 vs. 42/208; difference 23.4 points [14.4, 32.6]). For example, "Can you bring the dog?" becomes "I think you might bring the dog." The rewrite instruction itself asks for "a statement that is not sure", which a request or question cannot satisfy without changing type. This explains how the drift arises; it is not a randomized test of edit type, because the slots that receive each edit differ. Drift varies by model, from 13/110 (Qwen) to 44/102 (Gemma). Coverage is also not faithfulness: in the gallery's gain row, I2 adds a hedged reading of smell gas, but every suggestion invents a location, and slot 3 gains a negation the user did not ask about. An applicability-aware rewrite that may answer UNSUPPORTED is specified with its own kill rule but has not been run.[PENDING: T3]

Cost. I2 added 2.05 generator calls and 3.25 s per set (I2t: 1.95 calls); 412/428 rewrites passed the monitor's own check.

6.3 Chip: cheap, effective, and still not complete

Adding one chip raised critical-value match from 41.8% [37.1, 45.9] with the length-matched word to 94.8% [91.4, 97.6], a paired gain of 53.0 points [47.8, 58.2] (kill line: 10). That is 97.3% of the full-input level (97.4%), and all chips together reached 95.9% at 3.1 selections on average. Chip effects were near-complete for polarity and agent (100.0%, 100.0%) and weakest for speech act (81.2%) and politeness (90.0%).

Harm ledger. Adding the chip lost the critical value in 7/194 targets that the noncrit input had matched by chance, and 21/284 bare-input misses stayed wrong with the chip. The gallery shows one: with [?] bring dog, Llama produces the request "Will you bring the dog?" although the target is a question. No chip1 output contradicted a supplied dimen-

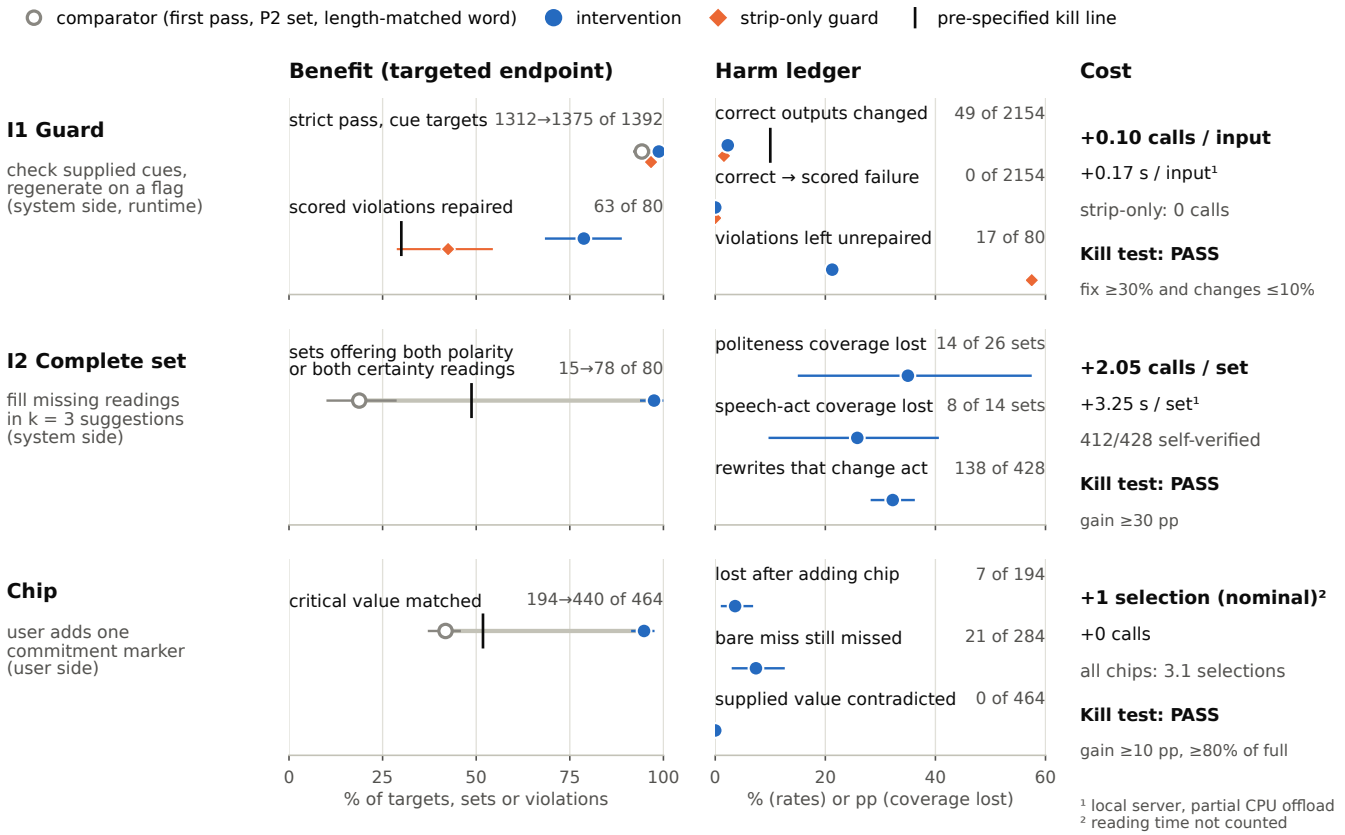


Figure 1: Benefit, harm and cost for each intervention on the same held-out benchmark (58 families, four models; machine-extracted labels). Benefit: the comparator (open circle) and the intervention (filled) on the targeted endpoint. Harm: the ledger entries; coverage losses are percentage points against the P2 set. Bars are 95% family-cluster bootstrap intervals. Black ticks are the kill lines fixed before the test run. Counts above each row give the observed numerator and denominator. Local latency is measured under partial CPU offload, and chip cost omits reading and navigation time.

sion (0/464), and unsupplied politeness was rarer with a chip (3.2%) than with the noncrit word (9.1%). The nominal cost of one selection is a lower bound: it omits finding the chip, reading the output and repairing it, and it shifts the burden of each non-default meaning onto the user (Frisch et al. 2026). A pre-registered replication on 240 fresh families (speech-act chip1 minus noncrit) is running.[PENDING: W-C] Whether adding a second chip damages a satisfied dimension is tested separately against repeat disagreement.[PENDING: W-D]

6.4 Heterogeneity across models

Treatment effects differ by system (Table 2). The guard matters most where the first pass is worst: Gemma starts lowest and the guard repairs 29/33 of its violations, while it repairs only 7/13 of Qwen's, the model with the fewest violations to begin with. I2's act drift ranges from 13/110 rewrites for Qwen to 44/102 for Gemma, so the same set intervention has a different harm profile on each backbone. Chip effects are nearly uniform. A deployment decision therefore needs the ledger for the specific model, and the human study (§7) should stratify on the backbone it ships.

	I1 guard			I2	Chip
Model	strict pass	repaired	changed	act drift	match
Qwen2.5-7B	96.3→98.3	7/13	3/335	13/110	42.2→95.7
Llama-3.1-8B	92.5→98.3	20/26	5/322	47/109	44.0→94.0
Gemma-3-4B	90.5→98.9	29/33	3/315	44/102	38.8→93.1
gpt-6.1-terra	97.7→99.7	7/8	3/340	34/107	42.2→96.6

Table 2: Per-model ledger. Strict pass on cue targets (%, P0→guard); violations repaired; already-correct cue targets whose text changed; I2 rewrites that change speech act; critical-value match (%, noncrit→chip1). Models are fixed systems, not a sample: the spread is descriptive.

What a gain-only report would have said. Three choices in the design change the headline. Without channel attribution, the guard would be credited with repairing 89.8% of all violations, including politeness added under ambiguous input, where no supplied commitment exists; on supplied cues it repairs 78.8%. Without the harm ledger, I2 is a 78.8-point gain. With it, the net gain over all six dimensions is 17.6

Arm	Visible input	Before (comparator)	After (intervention)	Model, item, declared rule, rate of that rule's population
Chip	like new medication → [NOT] like medication	"I like new medication"	"I don't like medication."	Qwen, pol-07, negated target. Rule: first negated polarity target (family, model order). Polarity match 50%→100% (all dims 194/464→440/464).
Chip	bring dog tomorrow → [?] bring dog	"I will bring the dog tomorrow."	"Will you bring the dog?"	Llama, act-07; target question, scored request. Rule: first speech-act chip failure. Speech-act chip match 81.2%; bare misses still missed with a chip: 21/284.
I1 repaired	[PLAIN] let finish sentence	"Please let me finish my sentence."	"Let me finish the sentence."	Gemma, pol2-14/A, chip1. Rule: first by hash order. 63/80 scored failures repaired.
I1 not repaired	[?] [YOU] charge wheelchair	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"	Llama, act-11/A, chipall. Rule: first by hash order. 17/80 failures remain; 12/17 of residuals involve speech act.
I1 repaired	[ASK] [YOU] [PLAIN] wear headphones	"You wear headphones."	"Ask you to wear headphones."	Llama, pol2-07/B, chipall. Rule: second by hash order. Scored pass, but not a usable sentence: 4/63 repairs start "Ask you" (post hoc lexical count; fluency unmeasured).
I1 changed correct	The user typed: Would you mind repeating the last question?	"Would you mind repeating the last question."	"Please repeat the last question."	Llama, pol2-13/A, full. Rule: first by hash order. Already-correct cue targets changed: 14/1312; none became a scored failure, yet the softener form changed.
I2 coverage gain	smell gas	1. "I smell gas in the kitchen." [plain] 2. "There is gas smell coming from the basement." [plain] 3. "Do you smell gas outside?" [?]	1. "I smell gas in the kitchen." [plain] 2. "There might be a gas smell coming from the basement."† [hedged] 3. "Do you not smell gas outside?"† [?]	Qwen, cer-11. Rule: first polarity/certainty set gaining both readings (hash order). Sets with both readings 15/80→78/80; 63/65 uncovered sets gained. Slot 3 also changed and locations were invented.
I2 collateral loss	knock come	1. "Someone knocked and came in." [plain] 2. "Did someone knock before coming in?" [plain] 3. "Please knock before you come in." [polite]	1. "Someone knocked and came in." [plain] 2. "Maybe someone knocked before coming in."† [plain] 3. "Don't knock before you come in."† [plain]	Terra, pol2-08. Rule: first politeness set losing both readings (hash order). Politeness coverage 26/40→12/40; 14/26 covered sets lost. Different families from the row above.
I2 act drift	bring dog	"Can you bring the dog?"	"I think you might bring the dog."	Llama, act-07, slot 2 (certainty rewrite). Rule: first certainty edit that changes act (family, model, slot). Certainty edits changing act: 96/220.

Figure 2: Before/after gallery of real stored outputs, with no edits. Each row is chosen by the declared rule it names (hash order is SHA-1 of request, family, variant and channel, as in the intervention report). Its rate describes the population of that rule, so the gallery is not a random sample. Set rows show all three suggestions in both arms, in slot order, each with its scored reading ([?] = unclear on that dimension); † marks slots I2 rewrote. Chip rows compare against the length-matched non-critical word. Labels are machine-extracted; human validation is pending.

points [4.0, 31.6], and 32.2% of its edits change sentence type. Without pairing on stored first passes, the guard's effect would be confounded with the 76.4–90.8% run-to-run reproducibility of greedy decoding on local models.

6.5 Reading the ledger across interventions

Fig. 1 places the three interventions on one page because each answers a different defect, and a single score would hide that. The guard enforces commitments the user supplied, at a small call cost, and its harms are small but real edits to correct outputs. The completed set exposes readings the input left open, at roughly twenty times the guard's call cost per

input, and its harms land on dimensions it did not target. The chip moves the work to the user and is the most effective per unit of system cost. Its harms are residual misses, and the burden of being the one who must mark every non-default meaning. For a designer the combinations matter: chips plus the guard enforce what was marked, and the copy baseline covers typed sentences. Under ambiguity, I2 trades politeness and sentence-type coverage for polarity and certainty.

7 From Offline Ledgers to Human Effects: A Protocol

The results above are effects on text. Whether they become effects on people is a causal and longitudinal question, and it may have a different answer. A guard that repairs most violations may also teach users to stop marking negation. A set that offers "maybe" may also make the speaker sound less certain over weeks of use. We specify the study that would answer this, and we use the offline ledger to design it.

Design. A micro-randomized trial (Klasnja et al. 2015) with AAC users who already compose with keyword expansion, over several weeks of their own use. A *decision point* is each expansion request. Randomization happens only where the arms differ, which the offline ledger identifies. *I1*: at decision points where the monitor flags the first pass (8.5% of inputs offline), randomize with probability 0.5 whether the repaired or the original output is shown. *I2*: at each bare-keyword suggestion request, randomize the P2 or I2 set (I2 changed nearly every set offline). *Chip*: chips are the user's choice and cannot be randomized directly; we randomize whether the chip bar is shown expanded or collapsed, an encouragement design that identifies the effect of chip availability. Logged randomization probabilities also allow off-policy evaluation of guard policies not deployed (e.g., repair only speech-act flags).

Outcomes and estimands. Proximal (per message): sent without edit; number of correction actions; time to send; and, on a small random sample of sent messages, a one-tap "said what I meant". Partner-side: repair turns in the following exchange, where logging is consented. Distal (per week): chip use per message, the share of plain vs. polite and hedged vs. unhedged messages sent, self-reported authorship and fatigue. The primary estimand is the causal excursion effect (Boruvka et al. 2018) of showing the intervention's output on the proximal outcome, averaged over decision points, with pre-specified moderators. Moderators are channel (cue supplied or not), dimension flagged, time in study (habituation), and recent correction burden (a time-varying state). Randomization at known probabilities protects these estimands from time-varying confounding by fatigue or conversation context.

Longitudinal and co-adaptation outcomes. Three offline findings become longitudinal hypotheses. (a) *Reliance*: if the guard repairs most violations, users may mark fewer commitments over time. The test is the chip-use trend, compared between users randomized to high vs. low guard exposure. (b) *Style drift*: the politeness additions and certainty rewrites seen offline predict a drift in the register of sent messages; we test it with the weekly plain/polite share, against the user's own pre-study baseline. (c) *Carry-over*: a partner who receives polite messages may respond differently (Zhang et al. 2026; Hohenstein et al. 2023), so we include lagged excursion effects on later decision points.

Harm-based stopping rules. The offline kill lines become human stopping rules, fixed in advance. The study stops an arm if "said what I meant" is lower under the intervention by more than a pre-registered margin, if changed-correct outputs are sent and then retracted above a threshold, or if participants turn the intervention off. The protocol is planned with AAC users and autistic advisers; it includes accessible consent, a visible off switch, local-only message logging, and paid participation (Janse van Rensburg et al. 2026).

What the offline ledger contributes. It shrinks the human study to decision points where the arms differ. It names the harms worth measuring before any user sees them. It supplies falsifiable surrogate hypotheses: if offline changed-correct predicts "said what I meant", the cheap ledger can screen later interventions. If it does not, offline evaluation of this kind should be demoted.

8 Limitations

No human outcomes. Nothing here measures communication success, agency, effort or trust. *Machine labels.* All endpoints use a blind LLM extractor and deterministic verdicts. The extraction pipeline passed identity and mutation tests (Anonymous 2027), but its pre-registered validation against independent human labels has no returns yet, so every rate is conditional on the extractor.[PENDING: human gold] Shared errors between monitor and extractor can inflate guard repairs; the planned repair audit targets this. *Constructed items.* Ten families per dimension and four models describe these systems and items, not AAC users' messages. A transfer study on fictional crowd-written AAC messages (Vertanen and Kristensson 2011) is in progress.[PENDING: W-B] *Narrow endpoints.* The labels cover six coarse dimensions; a scored pass can be a fragment, and added content (invented locations) is not penalized in coverage. *Selection.* I2 was chosen over I2t on development data; gallery rows are selected by declared rules and are not representative samples, so each carries its own rate. *Exploratory analyses.* The act-drift and residual decompositions were specified after earlier results were seen. They are descriptive, and their intervals are not corrected for that search. *Serving.* Intervention calls for local models ran with partial CPU offload; latency is comparable only within this study.

9 Conclusion

Each of the three interventions passed its pre-specified test. Each also has a measured cost that a gain-only report would leave out: edited correct outputs and unusable repairs for the guard, readings displaced in the completed set, and residual misses plus user burden for chips. Reporting benefit, harm and cost together turned a ranking question into a design

question that a human study can test. We release the analysis code, the stored outputs and the scripts that rebuild every number and figure in this paper from them.[PENDING: anonymized artifact URL]

References

Agarwal, D.; Naaman, M.; and Vashistha, A. 2025. AI Suggestions Homogenize Writing Toward Western Styles and Diminish Cultural Nuances. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. ACM.

Angelopoulos, A. N.; Bates, S.; Fannjiang, C.; Jordan, M. I.; and Zrnic, T. 2023. Prediction-powered inference. *Science*, 382(6671): 669–674.

Anonymous. 2027. Testing What an AAC Expander Is Allowed to Say: Input-Conditioned Commitment Contracts for LLM Sentence Expansion. Concurrent submission to SE4All at ICSE 2027; anonymized copy provided to reviewers.

Arnold, K. C.; Chauncey, K.; and Gajos, K. Z. 2020. Predictive Text Encourages Predictable Writing. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. ACM.

Boruvka, A.; Almirall, D.; Witkiewitz, K.; and Murphy, S. A. 2018. Assessing Time-Varying Causal Effect Moderation in Mobile Health. *Journal of the American Statistical Association*, 113(523): 1112–1121.

Cai, S.; Venugopalan, S.; Seaver, K.; Xiao, X.; Tomanek, K.; Morris, M. R.; Kane, S.; Narayanan, A.; MacDonald, R. L.; Kornman, E.; Vance, D.; Casey, B.; Gleason, S. M.; Nelson, P. Q.; and Brenner, M. P. 2024. Using large language models to accelerate communication for eye gaze typing users with ALS. *Nature Communications*, 15: 9449.

Choudhury, S. H.; Kumar, A.; and Martin, L. J. 2025. Evaluating Human-LLM Representation Alignment: A Case Study on Affective Sentence Generation for Augmentative and Alternative Communication. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, 1618–1637. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics.

Frisch, B.; Wade, W.; Gaines, D.; Kinsella, M.; Peters, B.; Broderick, T.; and Vertanen, K. 2026. It's Complicated: On the Design and Evaluation of AI-Powered AAC Interfaces. Position paper, Speech AI for All workshop at CHI 2026, arXiv:2606.24854.

Gorenshtein, A.; Omar, M.; Jia, E.; Adiniaev, Y.; Daniel, O.; Kruskal, J.; Ahmed, M.; Brook, O.; Klang, E.; and Barash, Y. 2026. Intent Drift in LLM-Assisted BCI Communication: An In-Silico Benchmark Under Simulated Decoder Corruption. MedRxiv preprint, posted 27 August 2026.

Gou, Z.; Shao, Z.; Gong, Y.; Shen, Y.; Yang, Y.; Duan, N.; and Chen, W. 2024. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing. In *International Conference on Learning Representations*.

Hancock, J. T.; Naaman, M.; and Levy, K. 2020. AI-Mediated Communication: Definition, Research Agenda, and Ethical Considerations. *Journal of Computer-Mediated Communication*, 25(1): 89–100.

Hohenstein, J.; Kizilcec, R. F.; DiFranzo, D.; Aghajari, Z.; Mieczkowski, H.; Levy, K.; Naaman, M.; Hancock, J.; and Jung, M. F. 2023. Artificial intelligence in communication impacts language and social relationships. *Scientific Reports*, 13: 5487.

Jakesch, M.; Bhat, A.; Buschek, D.; Zalmanson, L.; and Naaman, M. 2023. Co-Writing with Opinionated Language Models Affects Users' Views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Janse van Rensburg, M. G.; Niunin, E.; Ostrolenk, A.; Arsenault, S.; Wilson, A.; Weaver, C.; Courchesne, V.; and Hock, R. M. 2026. Understanding the Use of Community Advisory Boards in Autism Research: A Scoping Review. *Review Journal of Autism and Developmental Disorders*.

Klasnja, P.; Hekler, E. B.; Shiffman, S.; Boruvka, A.; Almirall, D.; Tewari, A.; and Murphy, S. A. 2015. Microrandomized Trials: An Experimental Design for Developing Just-in-Time Adaptive Interventions. *Health Psychology*, 34(Suppl.): 1220–1228.

Lacosse, E.; Tano, P.; and McNamee, D. C. 2026. Multi-System Cognitive Steering for Generative Communication. Preprint, version 1, posted 13 October 2026; manuscript identifies the BiAlign Workshop at CHI 2026, arXiv:2610.19177.

Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegreffe, S.; Alon, U.; Dziri, N.; Prabhumoye, S.; Yang, Y.; Gupta, S.; Majumder, B. P.; Hermann, K.; Welleck, S.; Yazdanbakhsh, A.; and Clark, P. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, volume 36.

Mao, L.; Lee, J. H.; Faroqi-Shah, Y.; and Valencia, S. 2025. Design Probes for AI-Driven AAC: Addressing Complex Communication Needs in Aphasia. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*, 404–422. ACM.

Padmakumar, V.; and He, H. 2024. Does Writing with Language Models Reduce Content Diversity? In *International Conference on Learning Representations*.

Panickssery, A.; Bowman, S. R.; and Feng, S. 2024. LLM Evaluators Recognize and Favor Their Own Generations. In *Advances in Neural Information Processing Systems*, volume 37.

Ribeiro, M. T.; Wu, T.; Guestrin, C.; and Singh, S. 2020. Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–4912. Association for Computational Linguistics.

Shen, J.; Yang, B.; Dudley, J.; and Kristensson, P. O. 2022. KWickChat: A Multi-Turn Dialogue System for AAC Using Context-Aware Sentence Generation by Bag-of-Keywords. In *Proceedings of the 27th International Conference on Intelligent User Interfaces*, 853–867. ACM.

Valencia, S.; Cave, R.; Kallarackal, K.; Seaver, K.; Terry, M.; and Kane, S. K. 2023. "The less I type, the better": How AI

Language Models can Enhance or Impede Communication for AAC Users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Huynh, J.; Jiang, E. Y.; Wu, Y.; Wan, T.; Zheng, Z.; Admoni, H.; Bigham, J. P.; and Pavel, A. 2024. COMPA: Using Conversation Context to Achieve Common Ground in AAC. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. ACM.

Valencia, S.; Pavel, A.; Santa Maria, J.; Yu, S.; Bigham, J. P.; and Admoni, H. 2020. Conversational Agency in Augmentative and Alternative Communication. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM.

Vertanen, K.; and Kristensson, P. O. 2011. The Imagination of Crowds: Conversational AAC Language Modeling using Crowdsourcing and Large Data Sources. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, 700–711.

Xiang, W.; Xu, Y.; Fang, Y.; Teng, Z.; Jiang, Z.; Hu, B.; and Yang, J. 2025. SocializeChat: A GPT-Based AAC Tool Grounded in Personal Memories to Support Social Communication. Accepted at IEEE SMC 2025 per author record, arXiv:2510.19017.

Zhang, Z. S.; Choi, Y.; Xiao, Z.; Liao, Q. V.; and Zhao, J. 2026. "Hi Alex" or "Dear Dr. Morgan"?: Exploring How AI-suggested Politeness Strategies Influence Email Writing and Social Perception Among Native and Non-native Speakers. *Proceedings of the ACM on Human-Computer Interaction*, 10(6). Article CSCW094.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36.